

Essex County Brownfield Revitalization Data Pipeline: An End-to-End Geospatial and CRM Integration

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Abstract

Effective community revitalization relies on the accessibility and accuracy of environmental data. This paper outlines the development of an end-to-end data engineering pipeline designed to process, geocode, and visualize environmental liabilities in Essex County, New Jersey. Utilizing raw dataset extractions from the New Jersey Department of Environmental Protection (NJDEP), the architecture integrates a relational CRM (Airtable), a custom Python-based geocoding engine (GeoPy/ArcGIS), geospatial mapping (ArcGIS Online), and digital outreach platforms (Constant Contact). The resulting system resolves critical data quality issues in legacy government records and provides a scalable framework for community systems management and stakeholder engagement.

1 Introduction

Brownfield sites - abandoned or underutilized properties complicated by the presence of hazardous substances - present significant challenges to urban redevelopment. The Center for Community Systems and similar organizations require highly accurate geospatial data to coordinate revitalization efforts.

The primary objective of this project is to transform raw, unstructured NJDEP "Known and Suspected Sites" reports into an actionable, interactive database. This pipeline automates the data cleaning process, accurately maps site coordinates, and establishes a communication framework to bridge the gap between environmental data and local stakeholders.

2 System Architecture and Methodology

The pipeline is divided into four distinct operational phases, moving from raw data extraction to community outreach execution.

2.1 Phase 1: Data Extraction and Relational Modeling

Raw data was sourced from the NJDEP DataMiner portal, specifically querying Essex County records. To manage this data effectively, a relational architecture was established using Airtable. The database schema consists of:

- **Sites Table:** Contains fields such as Site ID, Current Site Name, Address, Municipality, and Contamination Status.
- **Stakeholders Table:** Linked via foreign keys to assign community leaders, municipal officials, and environmental agents to specific remediation sites.

2.2 Phase 2: Data Engineering and Custom Geocoding

A significant challenge arose from the formatting of legacy government address data, which included non-standard ranges (e.g., “1007 1009 BROAD ST”) and appended municipal tags. Initial testing with open-source geocoders (OpenStreetMap/Nominatim) resulted in a 96% failure rate in coordinate extraction.

To resolve this, a custom Python script was engineered using the `pandas` and `geopy` libraries.

- **String Cleaning:** Applied regex and string manipulation to normalize addresses.
- **Algorithmic Geocoding:** Integrated the ArcGIS API via `geopy`, utilizing its superior parsing logic to bypass standard enterprise paywalls and achieve a 100% success rate in extracting precise Latitude and Longitude coordinates for all records.

2.3 Phase 3: Geospatial Visualization

The enriched, coordinate-mapped dataset was imported into ArcGIS Online. Utilizing a Light Gray Canvas basemap to maximize data contrast, interactive visual layers were created. The resulting dashboard successfully plotted sites across critical Essex County municipalities, including Newark, Irvington, and Maplewood, allowing for spatial cluster analysis of environmental liabilities.

2.4 Phase 4: Stakeholder Outreach and Engagement

To simulate the final phase of community systems management, the geospatial assets were integrated into Constant Contact. A professional email campaign template was designed to invite local stakeholders to a simulated Town Hall meeting. The outreach asset dynamically presented the ArcGIS visualizations, demonstrating how technical data can be packaged for public consumption and civic action.

3 Results and Impact

The implemented pipeline successfully ingested 50 unstructured environmental records and output a pristine, geocoded dataset. By migrating away from failing open-source geocoders to a targeted Python/ArcGIS API approach, the system achieved total spatial accuracy. Furthermore, the integration of Airtable ensures that the data is not static, but rather a living CRM capable of tracking ongoing community engagement.

4 Conclusion

This project demonstrates advanced proficiency in data engineering, API integration, and geospatial analytics. By bridging the gap between raw government data and community outreach platforms, this pipeline serves as a highly effective model for urban planning and environmental revitalization initiatives. The methodologies established here are highly scalable and directly applicable to the objectives of the Center for Community Systems.